Midterm Practice Questions · Answer Key

MATH60033A · Quantitative Methods in International Business

Thierry Warin, PhD · HEC Montréal · A2026

Open this only after answering the questions on paper. The answers are model answers: on the exam, any precise formulation in your own words earns the marks. In Parts I, II and D there is no single right answer — a precise concept and an argued position are what count.

Part I — Europe 2031

I1. *(easy)* A forecast says what will probably happen and can carry a probability; a scenario is a coherent, conditional story — *if* these things, *then* those — used to expose assumptions and prepare decisions, with no probability attached. The prologue (January 2025 – June 2026) draws on the observed record; from August 2026 the narrative is constructed. Anyone who quotes a post-seam event or number as evidence is citing a story as a fact.

I2. *(medium)* The file holds the model's learned parameters (weights) — billions of numbers set during training. When it answers, it computes, token by token, a probability for the next piece of text from the text so far, and picks one: pattern completion learned from its training data. Its competence comes from that training text; it stops where the text was thin, wrong or out of date (a library version released after training), and fluency is not understanding.

I3. *(medium)* They are internal parameters of the story — the stock of AI computing capacity, measured by the electrical power it draws — not forecasts or measurements. Choosing gigawatts assumes that power (energy and data centres) is the binding constraint on AI capability; it leaves out data, algorithmic efficiency, talent, adoption by firms and legitimacy.

I4. *(medium)* Compute: the chips and energy that train and run models. Data: what the models learn from — its quantity, quality, languages and legal access. Mathematics: the algorithms and model designs that turn compute and data into capability; better algorithms can substitute for compute. A good answer argues where Europe stands — e.g. weak on frontier compute, stronger on industrial and multilingual data, research and efficient models.

I5. *(medium)* AI built for one sector or task, trained on domain data, where value comes from know-how and data a firm holds rather than from frontier scale — e.g. predictive maintenance for machinery, quality control in manufacturing, logistics routing, medical imaging. Argued either way: yes, it plays to Europe's industrial strengths and needs less compute; or no, if general models keep improving they absorb vertical tasks and Europe still depends on them.

I6. *(medium)* A compact model, able to run locally or on modest hardware. Advantages: cost, data kept in-house (privacy, GDPR), independence from a foreign provider, lower energy, can be adapted to the firm's own task. Limits: less general capability, the firm must maintain and evaluate it, weaker on hard reasoning, still often trained abroad.

I7. *(medium)* Because in the scenario's logic capability comes from compute, and compute needs data centres, which need power, grid connections and permits — Europe's bottleneck. What could go wrong: energy and water use, grid strain, chips still imported (dependence moves rather than vanishes), and investing in a race Europe may not win while its advantage lies elsewhere.

I8. *(difficult)* No single right answer. Sovereignty: full control, but very costly and slow, and still dependent on foreign chips and tools. Governed interdependence: faster and cheaper, keeps access to frontier models, but leaves leverage with US firms and governments. A strong answer names the trade-off, takes a side, and says what would make the chosen path fail.

I9. *(difficult)* Sessions 2–5 are largely statistics — means, least squares, logistic regression (decades old), penalised regression (ridge from the 1970s, the lasso from 1996); what is new is scale, computing power and the use of the same ideas for prediction. The distinction matters because the old methods come with known assumptions and diagnostics — the questions of this course apply to "AI" just as well.

Part II — The readings

Pointers, not full answers. Credit any precise research question, a correct description of the model, two distinct critiques and a coherent alternative.

II1. *(medium)* **Fraiberger et al. (2021).** (a) Does a better-than-usual news day for a country go with higher stock returns in that country over the following days, and does global news matter more than local news? (b) Outcome: a country's stock returns over the following days. Unit: a country-day, for 25 countries. Main variable: a daily news-sentiment index built from Reuters articles (share of positive minus negative words, standardised within each country); controls for past returns, volume, volatility and global risk; a linear regression. (c) The index is a constructed measure that assumes word counts capture sentiment; news may react to markets rather than the reverse; a common cause may move both. (d) A logistic regression of whether the market rises or falls the next day, reporting odds ratios and predicted probabilities; or a lasso on many text-based measures to see which predict returns out of sample.

II2. *(medium)* **Amsili et al. (2024).** (a) Can cheap, routinely measured soil properties predict a soil's water-holding capacity well enough to replace direct measurement, and does a random forest beat a linear regression? (b) Outcome: field capacity (also wilting point and available water). Unit: a soil sample — 7,232 from the Cornell Soil Health Laboratory. Predictors: sand, clay, soil organic matter. Methods: a multiple linear regression against random forests; the regression explains about 79% of the variance, about as much as the reduced random forest on the same inputs. (c) The fit statistics are not computed the same way for the two methods (in-sample for the regression, out-of-bag for the forests); the samples are US topsoils sent to one lab, so the model may not travel to other soils; samples above 10% organic matter are excluded. (d) Ridge, lasso or elastic net with the other lab measurements as candidates, chosen by cross-validation and judged on held-out error; or a squared term if the relation is curved.

II3. *(difficult)* **Saganowski et al. (2019).** (a) Can a group's current structure in a social network predict which of six events — dissolve, shrink, grow, continue, merge, split — happens to it next? (b) Outcome: the next event, six unordered categories. Unit: a group in a time window. Predictors: about 90 network measures (size, density, centrality…). Methods: 15 classifiers compared; tree-based ones (bagging, random forest) do best, scored by the average of the per-class F-measures rather than accuracy, because accuracy rewards predicting the common events. (c) Strong class imbalance; many correlated features, so the feature ranking is unstable; results vary across data sets. (d) A multinomial logit — six unordered categories, one baseline — reporting relative risk ratios and predicted probabilities; with a lasso-type penalty to handle 90 correlated features.

II4. *(medium)* **Blonigen and Piger (2014).** (a) Of 56 candidate determinants of bilateral FDI, which few actually carry the signal? Hard because earlier studies each picked their own short list and disagree. (b) Outcome: bilateral FDI between a parent and a host country. Unit: a country pair. Candidates: 56 variables (gravity, cultural distance, trade agreements, income, skills, institutions, costs…). Method: Bayesian model averaging, which weights every combination of variables and gives each an inclusion probability. Robust: gravity, cultural distance, parent income, some trade agreements, host skill; not robust: host business costs, infrastructure, financial depth, institutions. (c) Association, not causation; correlated candidates share the credit, so a low inclusion probability is not "no effect"; results depend on the priors and on the candidate list. (d) The lasso or, with correlated candidates, the elastic net, with chosen by cross-validation: a short list of shrunken coefficients and its out-of-sample error.

II5. *(difficult)* **Topalova and Khandelwal (2011).** (a) Did India's tariff cuts of 1991 — on a firm's output and on its inputs — raise its productivity? (b) Outcome: a firm's productivity. Unit: a firm in a year — about 3,000 listed manufacturing firms followed over several years. Main variables: the tariffs on the firm's industry output and on its inputs; a linear regression that compares each firm with itself as its tariffs change. Both cuts raise productivity, the input channel more. (c) Tariff cuts might have targeted industries already improving (the paper argues the IMF-imposed reform rules this out); productivity is estimated, not observed; listed firms are not all firms. (d) A logistic regression of whether a firm starts exporting or exits after the reform, reporting odds ratios and probabilities. *(The design that follows the same firms over time is not examined on the midterm.)*

Part A — Multiple choice

Q	Answer	Level	Why
A1	**C**	medium	$\bar{x}$ is estimated from the same data, so the deviations sum to zero: only $n-1$ are free.
A2	**B**	easy	A trimmed mean drops a fixed share from each tail before averaging.

Q	Answer	Level	Why
A3	**A**	easy	The median is not pulled by the long right tail.
A4	**D**	medium	The rule holds for a roughly normal variable; symmetric is not enough.
A5	**C**	medium	Negative skewness: a long left tail pulls the mean below the median.
A6	**A**	easy	OLS minimises the sum of squared residuals.
A7	**D**	easy	Residual = observed − fitted.
A8	**B**	difficult	Frisch–Waugh–Lovell: a partial relationship, net of the other regressors.
A9	**C**	difficult	Constant variance is not needed for unbiasedness.
A10	**A**	easy	Never interpret an intercept you have no data near.
A11	**D**	medium	A VIF above about 10 signals near-collinearity: an unstable coefficient.
A12	**B**	medium	Leverage depends only on how unusual the regressor values are.
A13	**C**	easy	Investigate; report with and without; never delete silently.
A14	**A**	medium	BIC's penalty per parameter ($\ln n$) exceeds AIC's 2 once $n > 7$.
A15	**D**	easy	Likelihoods are only comparable on the same rows.
A16	**B**	medium	The omitted variable raises exports and moves with investment: the short regression gives investment its credit.
A17	**C**	easy	Without a design that isolates a cause, write "associated with".
A18	**A**	easy	$0.25/0.75 = 1/3$: one event for every three non-events.
A19	**D**	medium	$p = \text{odds}/(1 + \text{odds}) = 4/5 = 0.80$.
A20	**B**	easy	An odds ratio multiplies the odds, not the probability.
A21	**C**	easy	Logits are fitted by maximum likelihood.
A22	**A**	medium	Several coefficients jointly: a likelihood-ratio test of the nested models.

Q	Answer	Level	Why
A23	**D**	medium	McFadden's pseudo-R^2 compares log-likelihoods; 0.2–0.4 is already good.
A24	**B**	medium	The log compresses the tail; a coefficient on a logged outcome reads roughly as a % change.
A25	**C**	medium	J ordered categories: $J-1$ cutpoints, one slope per predictor.
A26	**A**	difficult	Proportional odds: the same odds ratio at every cut.
A27	**D**	difficult	The baseline changes what each coefficient compares, not the fitted probabilities.
A28	**B**	easy	Precision: of those flagged, the share truly positive.
A29	**C**	easy	Recall: of the true positives, the share flagged.
A30	**A**	medium	Calibration: predicted probabilities match observed frequencies.
A31	**D**	difficult	AUC measures ranking only; probabilities can be wrong while the ranking is right.
A32	**B**	medium	When misses are costly, flag at a lower probability.
A33	**C**	medium	Bias–variance trade-off: a little bias, much less variance.
A34	**A**	easy	A huge penalty shrinks every coefficient to zero: predict the mean.
A35	**D**	difficult	The penalty makes the problem solvable even with more predictors than rows.
A36	**B**	easy	The penalty is scale-dependent.
A37	**C**	medium	Leakage: the held-out folds influenced the scaling, so the error is optimistic.
A38	**A**	medium	The usual compromise between using enough data per fit and cost or instability.
A39	**D**	medium	Post-selection p-values are invalid.
A40	**B**	difficult	With more predictors than rows, the lasso keeps at most n variables.

Correct letters: A ×10, B ×10, C ×10, D ×10.

Part B — Short answers

B1. *(easy)* The mean is the sum divided by n, the balance point; the median is the middle value once sorted. The median is robust: an extreme value enters the sum and moves the mean, but not the middle position.

B2. *(easy)* $IQR = Q_3 - Q_1$, the width of the middle 50% of the data. It ignores both tails, so a long tail does not inflate it the way it inflates the standard deviation.

B3. *(easy)* A long **right** tail: a few large values, which pull the mean above the median.

B4. *(medium)* Not necessarily. The rule holds for a roughly **normal** variable; a symmetric distribution can still have heavy or light tails (check the excess kurtosis).

B5. *(easy)* The sum of squared residuals. Squaring makes every miss positive, so misses above and below cannot cancel, and punishes one large miss more than several small ones.

B6. *(medium)* "Comparing countries with the same investment, a one-percentage-point higher tertiary education share is associated with GDP growth 0.31 percentage points higher." Units, "holding investment fixed", and "associated with", not "causes".

B7. *(difficult)* The coefficient equals the slope from regressing the part of y not explained by the other regressors on the part of x_j not explained by them: what is left of the association once the other regressors have taken their share.

B8. *(easy)* The share of the variation in y that the model accounts for. It does not say the relationship is causal, nor that the model has the right shape, nor that it will predict new data.

B9. *(medium)* A **curve** (the straight line misses a non-linear relationship — try a squared term or a transformation); a **fan** (the spread of the errors changes with the fitted values); **isolated points** far from the rest (possible outliers or influential observations to investigate). Random scatter around zero is the model behaving.

B10. *(medium)* BIC, because its penalty per parameter, $\ln n$, exceeds AIC's 2 once $n > 7$. The models must be fitted on **the same rows**.

B11. *(easy)* It can predict probabilities below 0 or above 1, and it forces the effect on the probability to be the same everywhere, which is implausible near 0 and 1.

B12. *(easy)* Odds $= p/(1-p)$. For $p = 0.75$: $0.75/0.25 = 3$, "three events for every non-event".

B13. *(medium)* In **log-odds**: one more unit of x raises the log-odds by 0.26. 1.30 is the **odds ratio**: the odds are multiplied by 1.30, 30% higher. Wrong: "the probability is 30% higher".

B14. *(medium)* The probability is an S-shaped function of x: the same odds ratio moves the probability a lot where it is near 0.5 and little near 0 or 1, so the effect depends on where you start.

B15. *(medium)* Wald: one coefficient divided by its uncertainty, within one model. Likelihood ratio: compares the log-likelihoods of two nested models. Use the second for several coefficients **jointly**, such as all the dummies of one categorical variable.

B16. *(difficult)* The cutpoints are the thresholds between ordered categories, one per cut, acting as intercepts. Proportional odds: each predictor has **one** coefficient, so the same odds ratio applies at every cut. Check it by fitting a binary logit at each cut and comparing the slopes.

B17. *(difficult)* Discrimination: does the model **rank** positives above negatives (AUC)? Calibration: when it says 20%, does the event happen about 20% of the time? A model can rank perfectly while every probability is twice too high — useless for pricing or setting thresholds.

B18. *(medium)* It makes one in-or-out decision at a time, each depending on the previous ones; resampling the rows changes the path and the final model, and noise variables get in. The selection reflects this sample, not a property of the world.

B19. *(medium)* It sets weak coefficients to **exactly zero** (its absolute-value penalty), so it selects variables. Limits: with more predictors than rows it keeps at most n; among correlated predictors it keeps one, almost arbitrarily; the selected set is not "the true model".

B20. *(difficult)* Standardising on all the rows lets information from the held-out fold enter the training; the held-out fold is then not truly unseen, and the cross-validated error is too optimistic (leakage).

B21. *(medium)* With a squared term the effect of x is $b_1 + 2b_2x$: it depends on x, and b_1 is the effect only at $x = 0$. A negative b_2 means the effect weakens as x grows, and can turn negative past a turning point.

Part C — Reading output

C1. *(medium)* The empirical rule is applied to a variable that is clearly skewed: income cannot be negative, and a standard deviation of 71% of the mean signals a long right tail. Report the median and the IQR instead.

C2. *(easy)* "Not significant" is read as "no effect". With $p = 0.31$ the data cannot distinguish the effect from zero, but large negative effects are not ruled out. Say the evidence is inconclusive.

C3. *(medium)* AIC compared on different rows, 1,200 against 900: the likelihoods are not comparable, and most of the drop comes from having fewer observations. Refit both on the same 900 rows.

C4. *(medium)* An odds ratio read as a ratio of probabilities. 1.40 means the **odds** are 40% higher; the change in probability depends on the starting point. (And collateral may reflect which borrowers are asked for it — not a cause.)

C5. *(medium)* Accuracy on a 1% outcome: predicting "never fraud" is 99% accurate and catches nothing. Report the base rate, precision and recall. And with misses fifty times as costly, the threshold should be far below 0.5.

C6. *(difficult)* Three errors. Standardising on **all** rows before cross-validation (leakage). Treating the lasso's selection as the true model. Reporting OLS p-values on the selected columns: the same data chose and then tested them. Fix: put the scaler inside a pipeline, report the selected set and its out-of-sample error, no post-selection p-values.

C7. *(medium)* A high R^2 does not show the shape is right: the U-shaped residuals say the straight line misses a curvature. Add a squared term (or transform the variables) and check the residuals again.

C8. *(difficult)* A multinomial coefficient compares *general* with the **baseline** (academic), not the probability of general itself. Every equation moves at once: here better writers leave the vocational programme faster than the general one, so $P(\text{general})$ first rises. Report the predicted probabilities, and say "relative to academic".

Part D — Judgement (what a strong answer contains)

D1. *(medium)* **What it is:** a cross-country association: countries with more schooling export more high-technology goods. **Not licensed:** "will raise" — richer countries both school longer and export more technology (reverse causality and common causes: income, institutions, foreign investment); R^2 says nothing about cause; and the effect of *compulsory* schooling on those who would otherwise leave school may differ from the average difference between countries. **Evidence needed:** a design that isolates the change — countries that extended schooling compared with similar ones that did not, before and after, with the delay it takes for pupils to reach the labour market. **Meanwhile:** report the association with its uncertainty, attach no export figure to the policy, and plan an evaluation.

D2. *(difficult)* **What it is:** a predictive association in this bank's data. **Not licensed:** region may stand in for income, sector or collateral; refusing a whole region also raises fairness and legal issues. **Evidence:** does the odds ratio survive the borrower's own risk factors? Is the model calibrated for that region? **Meanwhile:** price risk on the borrower's own characteristics, use a cost-based threshold, and review whether region is a legitimate variable.

D3. *(medium)* **What it is:** six variables that jointly predict well at one λ, on this sample. **Not licensed:** a dropped variable is not "no effect" — the lasso keeps one of a correlated group almost arbitrarily. **Evidence:** how often each variable is kept across resamples; an elastic net to see whether groups enter together; a causal design for any variable whose effect matters. **Meanwhile:** report predictive performance, not a list of "drivers".

D4. *(difficult)* **What it is:** an AUC of 0.95 says the model **ranks** borrowers well. **Not licensed:** pricing needs the probabilities themselves to be right, and AUC says nothing about calibration — the probabilities could all be twice too high or too low. **Evidence:** a calibration check on recent loans (predicted against observed default rates by group), on data the vendor did not train on. **Meanwhile:** use the score to rank or screen, not to price, until it is calibrated.

D5. *(easy)* The mean is pulled up by a few very large firms: a skewness of 6.4 is far beyond any threshold, and the median (18) is a fourteenth of the mean. The typical firm has about 18 employees — small, not medium. Report the median and the IQR, and say the distribution is strongly right-skewed.

Questions: `PRACTICE.md` · *the exam:* `EXAM.md` *and its solutions* · *the summary:* `MIDTERM-SUMMARY.pdf`